Partial Differentiation, Explained Simply

Engineering Mathematics – I (UE26MA141A) · Unit 1

A beginner-friendly companion to the assignment solutions

Written so that a high-school student who knows basic derivatives can follow every line.

- **Part 0** – Warm-up: what a derivative really is (and the rules you need)
- **Part 1** – Partial derivatives (Problems 1–8)
- **Part 2** – Chain rule, total derivative, implicit functions (Problems 9–16)
- **Part 3** – Homogeneous functions and Euler's theorem (Problems 17–22)
- **Part 4** – Taylor and Maclaurin series (Problems 23–26)
- **Part 5** – Maxima and minima (Problems 27–28)
- **Part 6** – Lagrange multipliers (Problems 29–30)

How to read this booklet: each part starts with the *idea* (the picture in your head), then the *tool* (the formula), then works each problem slowly, saying **why** each step is done.

Colour key: Big idea Why this step? In plain words Watch out Takeaway

Part 0 — Warm-up: What a Derivative Really Is

0.1 The one-sentence meaning

A derivative answers one question: **"If I nudge the input a tiny bit, how fast does the output change?"**

If you drive a car and $s(t)$ is the distance travelled after t seconds, then $\frac{ds}{dt}$ is your speed. Distance is the output, time is the input, and the derivative measures how quickly the output reacts to the input.

On a graph, the derivative at a point is the **slope of the tangent line** there: how steep the curve is at that exact spot.

> **In plain words**
> $\frac{dy}{dx}$ = "change in y per unit change in x, when the change is made extremely small." If $\frac{dy}{dx} = 3$, a tiny step of 0.01 in x makes y go up by about 0.03.

0.2 The rules you will use all the time

Rule	Formula	What it means
Constant	$\frac{d}{dx}(c) = 0$	Something that never changes has zero rate of change.
Power	$\frac{d}{dx}x^n = nx^{n-1}$	Bring the power down, reduce it by one.
Constant multiple	$\frac{d}{dx}(cf) = cf'$	Constants just ride along.
Product	$(fg)' = f'g + fg'$	Differentiate one at a time, keep the other.
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$	"Low d-high minus high d-low, over low squared."
Chain	$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x)$	Outside derivative × inside derivative.

Derivatives of standard functions

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
$\sin x$	$\cos x$	e^x	e^x
$\cos x$	$-\sin x$	a^x	$a^x \log a$
$\tan x$	$\sec^2 x$	$\log x$ (natural log)	$\frac{1}{x}$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$	$\cot x$	$-\csc^2 x$

Throughout this booklet log means the natural logarithm (base e), the same as ln.

Why the chain rule matters so much here

Almost every problem in this unit is a "function inside a function" (like $\tan^{-1}$ of a fraction, or log of a polynomial). The chain rule is how we peel those layers: differentiate the outer layer, then multiply by the derivative of what was inside. Example: $\frac{d}{dx}\log(x^2+1) = \frac{1}{x^2+1}\cdot 2x$. The $\frac{1}{x^2+1}$ comes from "log of something", the $2x$ from "the something".

Part 1 — Partial Derivatives

1.1 Functions with more than one input

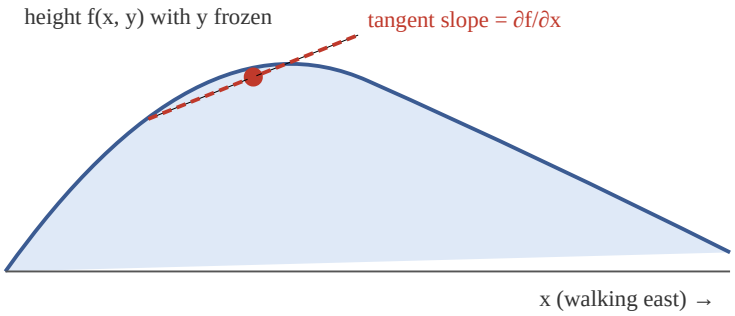
So far you have seen functions with one input, like $y = x^2$. But many real quantities depend on several things at once:

- The **area** of a rectangle depends on its length x and width y: $A = xy$.
- The **height** of a hill depends on how far east (x) and how far north (y) you stand: $h(x, y)$.
- The **temperature** in a room depends on position (x, y, z) and time t.

We write such a function as $u = f(x, y)$ or $u(x, y, z)$, and the graph of $f(x, y)$ is a **surface** (like a landscape) instead of a curve.

1.2 The idea of a partial derivative

> **Big idea**
>
> Imagine standing on a hill. The steepness depends on *which direction* you walk. A **partial derivative** picks one direction and measures steepness only along it:
>
> - $\dfrac{\partial f}{\partial x}$: slope if you walk **due east** (only x changes, y stays fixed).
> - $\dfrac{\partial f}{\partial y}$: slope if you walk **due north** (only y changes, x stays fixed).



Freeze y, and the surface becomes an ordinary curve in x. The partial derivative $\partial f/\partial x$ is the slope of that curve.

> **The golden rule (memorise this)**
>
> To find $\dfrac{\partial f}{\partial x}$: **pretend every other letter is just a number** (a constant) and differentiate as usual in x.
>
> To find $\dfrac{\partial f}{\partial y}$: pretend x (and every other letter) is a constant.

Example. $f = x^2y^3 + 5y$.

- $\dfrac{\partial f}{\partial x}$: treat y like the number 7, say. Then $f = x^2 \cdot (\text{const}) + (\text{const})$, so $\dfrac{\partial f}{\partial x} = 2x\,y^3 + 0 = 2xy^3$.
- $\dfrac{\partial f}{\partial y}$: treat x as constant: $\dfrac{\partial f}{\partial y} = x^2 \cdot 3y^2 + 5 = 3x^2y^2 + 5$.

Notation. The curly ∂ ("del" or "partial") reminds you that other variables exist and are being held fixed. Short forms: $f_x = \dfrac{\partial f}{\partial x}$, $f_y = \dfrac{\partial f}{\partial y}$.

1.3 Second-order and mixed partial derivatives

A partial derivative is itself a function of x and y, so we can differentiate it again:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \qquad f_{xy} = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

f_{xx} tells you how the east-slope itself changes as you walk east, which is the **curvature** (bending) in that direction. f_{xy} is a **mixed** derivative: how the east-slope changes when you step north.

Clairaut's theorem (order doesn't matter)

For all the smooth functions in this course, $\dfrac{\partial^2 f}{\partial x\,\partial y} = \dfrac{\partial^2 f}{\partial y\,\partial x}$. Differentiating first by x then y gives the same answer as y then x. Problem 3 checks this on an example.

Problem 1. Find the first-order partial derivatives of (i) $u = \tan^{-1}\dfrac{x^2+y^2}{x+y}$, (ii) $u = \cos^{-1}\dfrac{x}{y}$.

Goal: find $\dfrac{\partial u}{\partial x}$ and $\dfrac{\partial u}{\partial y}$. "First order" means differentiate once.

Game plan

Both are "inverse-trig of a fraction". So it is a chain rule problem: derivative of the outer inverse-trig function, times the partial derivative of the inner fraction.

Part (i)

Step 1: Give the inside a name. Let $w = \dfrac{x^2+y^2}{x+y}$, so $u = \tan^{-1} w$.

Why?

Naming the messy inner part keeps the work tidy. We only have to remember $\dfrac{d}{dw}\tan^{-1} w = \dfrac{1}{1+w^2}$, and the chain rule says $\dfrac{\partial u}{\partial x} = \dfrac{1}{1+w^2}\cdot\dfrac{\partial w}{\partial x}$.

Step 2: Differentiate the inside with respect to x using the quotient rule, holding y fixed:

$$\frac{\partial w}{\partial x} = \frac{(\text{top})'\cdot(\text{bottom}) - (\text{top})\cdot(\text{bottom})'}{(\text{bottom})^2} = \frac{2x\,(x+y) - (x^2+y^2)\cdot 1}{(x+y)^2}.$$

The derivative of the top x^2+y^2 w.r.t. x is $2x$ (the y^2 is a constant, so it gives 0). The derivative of the bottom $x+y$ is 1.

Simplify the numerator: $2x^2+2xy-x^2-y^2 = x^2+2xy-y^2$. So $\dfrac{\partial w}{\partial x} = \dfrac{x^2+2xy-y^2}{(x+y)^2}$.

Step 3: Simplify $1+w^2$.

$$1 + w^2 = 1 + \frac{(x^2+y^2)^2}{(x+y)^2} = \frac{(x+y)^2 + (x^2+y^2)^2}{(x+y)^2}.$$

We wrote 1 as $\dfrac{(x+y)^2}{(x+y)^2}$ to get a common denominator.

Step 4: Multiply.

$$\frac{\partial u}{\partial x} = \frac{1}{1+w^2}\cdot\frac{\partial w}{\partial x} = \frac{(x+y)^2}{(x+y)^2+(x^2+y^2)^2}\cdot\frac{x^2+2xy-y^2}{(x+y)^2} = \frac{x^2+2xy-y^2}{(x+y)^2+(x^2+y^2)^2}.$$

The $(x+y)^2$ cancels, which is the payoff of writing $1+w^2$ as a single fraction.

Step 5: Use symmetry for $\partial u/\partial y$. If you swap x and y in u, you get exactly the same u (since x^2+y^2 and $x+y$ don't change). So the answer for $\partial u/\partial y$ is the x-answer with x and y swapped:

$$\frac{\partial u}{\partial y} = \frac{y^2 + 2xy - x^2}{(x+y)^2 + (x^2+y^2)^2}.$$

Why symmetry works

A symmetric function treats x and y the same way, so "how it reacts to y" must look like "how it reacts to x" with the letters exchanged. This is a useful shortcut that saves you redoing the calculation.

Part (ii)

Step 1: Recall the outer derivative. $\dfrac{d}{dt}\cos^{-1} t = -\dfrac{1}{\sqrt{1-t^2}}$, with $t = \dfrac{x}{y}$.

Step 2: Simplify the square root. $\sqrt{1 - \dfrac{x^2}{y^2}} = \sqrt{\dfrac{y^2 - x^2}{y^2}} = \dfrac{\sqrt{y^2 - x^2}}{y}$ (taking $y > 0$). So $-\dfrac{1}{\sqrt{1-t^2}} = -\dfrac{y}{\sqrt{y^2 - x^2}}$.

Step 3: Derivative of the inside. $\dfrac{\partial}{\partial x}\left(\dfrac{x}{y}\right) = \dfrac{1}{y}$ (y is a constant, so x/y is just x times the constant $1/y$).

$\dfrac{\partial}{\partial y}\left(\dfrac{x}{y}\right) = x \cdot \left(-\dfrac{1}{y^2}\right) = -\dfrac{x}{y^2}$ (write $x/y = x\,y^{-1}$ and use the power rule).

Step 4: Multiply outer × inner.

$$\frac{\partial u}{\partial x} = -\frac{y}{\sqrt{y^2 - x^2}} \cdot \frac{1}{y} = -\frac{1}{\sqrt{y^2 - x^2}}, \qquad \frac{\partial u}{\partial y} = -\frac{y}{\sqrt{y^2 - x^2}} \cdot \left(-\frac{x}{y^2}\right) = \frac{x}{y\sqrt{y^2 - x^2}}.$$

Watch out

Don't forget the minus sign in the derivative of $\cos^{-1}$. In $\partial u/\partial y$, two minus signs multiply to give a plus.

Takeaway

Partial differentiation is ordinary differentiation with the other letters frozen. Name the inside, apply the chain rule, simplify.

Problem 2. If $u = x^y$, show that $\dfrac{x}{y}\dfrac{\partial u}{\partial x} + \dfrac{1}{\log x}\dfrac{\partial u}{\partial y} = 2u$.

Game plan

x^y is special: it looks like a *power* when y is fixed, and like an *exponential* when x is fixed. So the two partial derivatives use two different rules. Then we just substitute and simplify.

Step 1: $\partial u/\partial x$. Freeze y. Then x^y is like x^5: power rule gives $\dfrac{\partial u}{\partial x} = y\,x^{y-1}$.

Step 2: $\partial u/\partial y$. Freeze x. Then x^y is like 2^y: exponential rule $\dfrac{d}{dy}a^y = a^y \log a$ gives $\dfrac{\partial u}{\partial y} = x^y \log x$.

Watch out

A very common mistake is to write $\dfrac{\partial}{\partial y}x^y = y\,x^{y-1}$. That's wrong: the variable is in the *exponent*, so the power rule doesn't apply.

Step 3: Substitute into the left side.

$$\frac{x}{y} \cdot y\,x^{y-1} = x \cdot x^{y-1} = x^y, \qquad \frac{1}{\log x} \cdot x^y \log x = x^y.$$

The y cancels in the first term; $\log x$ cancels in the second. (Laws of exponents: $x \cdot x^{y-1} = x^{1+y-1} = x^y$.)

Step 4: Sum $= x^y + x^y = 2x^y = 2u$. Done.

Takeaway

Always ask "where is the variable I'm differentiating?" In the base ⇒ power rule. In the exponent ⇒ exponential rule.

Problem 3. Verify that $\dfrac{\partial^2 u}{\partial x\,\partial y} = \dfrac{\partial^2 u}{\partial y\,\partial x}$ for $u = x^3 + y^3 + 3axy$.

Game plan

"Verify" means: calculate both sides separately and see they come out equal. We are checking Clairaut's theorem (section 1.3) on an example. a is just a fixed number.

Step 1: First derivatives.

$\dfrac{\partial u}{\partial x} = 3x^2 + 0 + 3ay$ (y^3 is constant; $3axy$ is $(3ay)\cdot x$, whose derivative is $3ay$).

$\dfrac{\partial u}{\partial y} = 0 + 3y^2 + 3ax$.

Step 2: Differentiate the other way.

Take $\dfrac{\partial u}{\partial x} = 3x^2 + 3ay$ and differentiate w.r.t. y: $3x^2$ is constant (gives 0), $3ay$ gives $3a$. Result: $3a$.

Take $\dfrac{\partial u}{\partial y} = 3y^2 + 3ax$ and differentiate w.r.t. x: result $3a$.

Step 3: Both equal $3a$, so the mixed partials agree. Verified.

Why should they agree?

$\partial^2 u/\partial y\partial x$ asks "how does the east-slope change as I go north?" and $\partial^2 u/\partial x\partial y$ asks "how does the north-slope change as I go east?". Both measure the same "twist" of the surface, viewed from two sides.

Problem 4. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show (i) $\left(\dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}\right)^2 u = -\dfrac{9}{(x+y+z)^2}$, (ii) the expanded version of the same thing.

What does $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u$ mean?

Think of $D = \dfrac{\partial}{\partial x} + \dfrac{\partial}{\partial y} + \dfrac{\partial}{\partial z}$ as a **machine**: you feed it a function and it returns $u_x + u_y + u_z$. "$D^2 u$" means: run the machine twice, i.e. $D(Du)$.

Step 1: The three first derivatives. Using $\dfrac{\partial}{\partial x}\log(\text{stuff}) = \dfrac{1}{\text{stuff}}\cdot\dfrac{\partial(\text{stuff})}{\partial x}$:

$$u_x = \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz}, \quad u_y = \frac{3y^2 - 3xz}{\cdots}, \quad u_z = \frac{3z^2 - 3xy}{\cdots}.$$

(For u_x: $x^3 \to 3x^2$; $y^3, z^3 \to 0$; $-3xyz \to -3yz$ because yz is a constant.)

Step 2: Add them. Same denominator, so add tops:

$$Du = u_x + u_y + u_z = \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{x^3 + y^3 + z^3 - 3xyz}.$$

Step 3: Use a famous factorisation.

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx).$$

(You can check by multiplying out.) The bracket on top cancels with the bracket on the bottom:

$$Du = \frac{3}{x+y+z}.$$

Why look for this factorisation?

The numerator looked very similar to a piece of the denominator. Whenever that happens, check whether the denominator factors, because cancelling turns a scary expression into a tiny one.

Step 4: Run the machine a second time. $D^2u = D\left(\dfrac{3}{x+y+z}\right)$. Write $\dfrac{3}{x+y+z} = 3(x+y+z)^{-1}$. Its x-derivative is $-3(x+y+z)^{-2}\cdot 1$, and the y- and z-derivatives are the same. Adding three equal pieces:

$$D^2u = -\frac{3}{(x+y+z)^2}\times 3 = -\frac{9}{(x+y+z)^2}. \quad \text{That is (i).}$$

Step 5: Part (ii). Expand D^2 like squaring $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$:

$$D^2 = \frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+2\frac{\partial^2}{\partial x\partial y}+2\frac{\partial^2}{\partial y\partial z}+2\frac{\partial^2}{\partial z\partial x}.$$

(The cross terms pair up, e.g. $\partial_x\partial_y+\partial_y\partial_x = 2\partial_x\partial_y$, because mixed partials are equal by Clairaut.) So the long expression in (ii) *is* D^2u, which we already found.

Takeaway

Simplify after the first derivative *before* differentiating again. Doing the second derivatives of the original log directly would be a huge mess.

Problem 5. If $\theta = t^n e^{-r^2/(4t)}$, find n so that $\dfrac{1}{r^2}\dfrac{\partial}{\partial r}\left(r^2\dfrac{\partial\theta}{\partial r}\right) = \dfrac{\partial\theta}{\partial t}$.

What is going on?

This equation is the **heat equation** (in a sphere-shaped setting): θ is temperature, r is distance from the centre, t is time. We are asked: for which power n does this formula actually describe heat spreading out? Strategy: compute the left side, compute the right side, set them equal, and solve for n.

Step 1: $\partial\theta/\partial r$. Here t is constant, so t^n is just a number in front. Chain rule on the exponential:

$$\frac{\partial}{\partial r}e^{-r^2/4t} = e^{-r^2/4t}\cdot\left(-\frac{2r}{4t}\right) = -\frac{r}{2t}e^{-r^2/4t}.$$

So $\dfrac{\partial\theta}{\partial r} = -\dfrac{r}{2}\,t^{n-1}e^{-r^2/4t}$ (since $t^n/t = t^{n-1}$).

Step 2: Multiply by r^2. $r^2\dfrac{\partial\theta}{\partial r} = -\dfrac{1}{2}t^{n-1}\,r^3e^{-r^2/4t}$.

Step 3: Differentiate that w.r.t. r (product rule on $r^3\cdot e^{-r^2/4t}$):

$$\frac{\partial}{\partial r}\left(r^3e^{-r^2/4t}\right) = 3r^2e^{-r^2/4t}+r^3\cdot\left(-\frac{r}{2t}\right)e^{-r^2/4t} = \left(3r^2-\frac{r^4}{2t}\right)e^{-r^2/4t}.$$

Multiply by $-\frac{1}{2}t^{n-1}$ and divide by r^2:

$$\text{LHS} = t^{n-1}e^{-r^2/4t}\left(-\frac{3}{2}+\frac{r^2}{4t}\right).$$

Step 4: The right side, $\partial\theta/\partial t$. Now r is constant and both factors contain t, so use the product rule:

$$\frac{\partial\theta}{\partial t} = nt^{n-1}e^{-r^2/4t}+t^n\cdot e^{-r^2/4t}\cdot\frac{r^2}{4t^2}.$$

(The derivative of $-\dfrac{r^2}{4t} = -\dfrac{r^2}{4}t^{-1}$ is $+\dfrac{r^2}{4}t^{-2}$.) Factor out $t^{n-1}e^{-r^2/4t}$: $\text{RHS} = t^{n-1}e^{-r^2/4t}\left(n + \dfrac{r^2}{4t}\right)$.

Step 5: Set LHS = RHS. The common factor $t^{n-1}e^{-r^2/4t}$ is never zero, so divide it away:

$$-\frac{3}{2} + \frac{r^2}{4t} = n + \frac{r^2}{4t} \;\Rightarrow\; n = -\frac{3}{2}.$$

Why factor out the same thing on both sides?

Writing both sides as (same factor) × (simple bracket) lets us compare the brackets directly. Then the $r^2/4t$ terms cancel, leaving one equation in n.

Takeaway

"Find the value that makes an equation true" = compute both sides, simplify to a common form, compare.

Problem 6. If $\dfrac{x^2}{a^2+u} + \dfrac{y^2}{b^2+u} + \dfrac{z^2}{c^2+u} = 1$, prove $u_x^2 + u_y^2 + u_z^2 = 2(xu_x + yu_y + zu_z)$.

The twist: u is hidden

We are not told $u = \ldots$ directly. Instead u is defined **implicitly**: for each (x, y, z), u is whatever number makes the equation true. We can't easily solve for u, but we don't need to. We can differentiate the whole equation, remembering that u secretly depends on x, y, z.

Step 1: Differentiate both sides w.r.t. x. The right side is 1, whose derivative is 0. On the left, every term contains u, and u changes with x:

- $\dfrac{\partial}{\partial x}\dfrac{x^2}{a^2+u} = \dfrac{2x}{a^2+u} - \dfrac{x^2}{(a^2+u)^2}u_x$ (product/quotient rule; the second piece is because the denominator changes through u).
- $\dfrac{\partial}{\partial x}\dfrac{y^2}{b^2+u} = -\dfrac{y^2}{(b^2+u)^2}u_x$ (y is constant, but u is not!)
- Similarly $-\dfrac{z^2}{(c^2+u)^2}u_x$.

So $\dfrac{2x}{a^2+u} - V\,u_x = 0$, where $V = \dfrac{x^2}{(a^2+u)^2} + \dfrac{y^2}{(b^2+u)^2} + \dfrac{z^2}{(c^2+u)^2}$.

Step 2: Solve for u_x, and by the same reasoning u_y, u_z:

$$u_x = \frac{2x}{(a^2+u)V}, \qquad u_y = \frac{2y}{(b^2+u)V}, \qquad u_z = \frac{2z}{(c^2+u)V}.$$

Why give the long bracket the name V?

It appears in all three derivatives. Naming it lets us spot that it shows up again in the next step, and cancels.

Step 3: Left side. Square and add:

$$u_x^2 + u_y^2 + u_z^2 = \frac{4}{V^2}\left[\frac{x^2}{(a^2+u)^2} + \frac{y^2}{(b^2+u)^2} + \frac{z^2}{(c^2+u)^2}\right] = \frac{4}{V^2}\cdot V = \frac{4}{V}.$$

The bracket is exactly V again.

Step 4: Right side.

$$2(xu_x + yu_y + zu_z) = \frac{4}{V}\left[\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u}\right] = \frac{4}{V}\cdot 1 = \frac{4}{V}.$$

The bracket is the original equation, which equals 1. Both sides are $\dfrac{4}{V}$. Proved.

Takeaway

When a variable is defined by an equation, differentiate the equation itself. Then use the original equation again at the end; it often simplifies things to 1.

Problem 7. If $x = r\cos\theta$, $y = r\sin\theta$, show $\dfrac{\partial^2 r}{\partial x^2} + \dfrac{\partial^2 r}{\partial y^2} = \dfrac{1}{r}$.

Picture

(r, θ) are **polar coordinates**: r is the distance from the origin to the point (x, y), and θ is the angle. Here we need r as a function of x and y. By Pythagoras, $r^2 = x^2 + y^2$.

Step 1: First derivatives. Differentiate $r^2 = x^2 + y^2$ w.r.t. x (remember r depends on x): $2r\,r_x = 2x \Rightarrow r_x = \dfrac{x}{r}$. Likewise $r_y = \dfrac{y}{r}$.

Why not use $r = \sqrt{x^2 + y^2}$ directly?

You can, and you get the same answer. Differentiating $r^2 = x^2 + y^2$ is just quicker and avoids square roots.

Step 2: Second derivative. Differentiate $r_x = \dfrac{x}{r}$ w.r.t. x with the quotient rule. Top x has derivative 1; bottom r has derivative $r_x = \dfrac{x}{r}$:

$$r_{xx} = \frac{1 \cdot r - x \cdot \frac{x}{r}}{r^2} = \frac{r^2 - x^2}{r^3} = \frac{y^2}{r^3}.$$

(Multiply top and bottom by r; then use $r^2 - x^2 = y^2$.) By symmetry, $r_{yy} = \dfrac{x^2}{r^3}$.

Step 3: Add. $r_{xx} + r_{yy} = \dfrac{x^2 + y^2}{r^3} = \dfrac{r^2}{r^3} = \dfrac{1}{r}$. Done.

Problem 8. If $u = \dfrac{1}{\sqrt{x^2 + y^2 + z^2}}$, prove $u_{xx} + u_{yy} + u_{zz} = 0$.

Why this is famous

The expression $u_{xx} + u_{yy} + u_{zz}$ is called the **Laplacian**. Functions where it equals 0 are called **harmonic**. The function $1/r$ is the shape of the gravitational and electric potential of a point mass or charge. So this problem shows why gravity and electrostatics obey "Laplace's equation" in empty space.

Step 1: Simplify notation. Let $r = \sqrt{x^2 + y^2 + z^2}$, so $u = r^{-1}$, and (as in Problem 7) $r_x = \dfrac{x}{r}$.

Step 2: First derivative. Chain rule: $u_x = -r^{-2} \cdot r_x = -r^{-2} \cdot \dfrac{x}{r} = -x\,r^{-3}$.

Step 3: Second derivative. Product rule on $-x \cdot r^{-3}$:

$$u_{xx} = -1 \cdot r^{-3} + (-x) \cdot (-3r^{-4}) \cdot \frac{x}{r} = -\frac{1}{r^3} + \frac{3x^2}{r^5}.$$

Similarly $u_{yy} = -\dfrac{1}{r^3} + \dfrac{3y^2}{r^5}$ and $u_{zz} = -\dfrac{1}{r^3} + \dfrac{3z^2}{r^5}$.

Step 4: Add. $-\dfrac{3}{r^3} + \dfrac{3(x^2 + y^2 + z^2)}{r^5} = -\dfrac{3}{r^3} + \dfrac{3r^2}{r^5} = -\dfrac{3}{r^3} + \dfrac{3}{r^3} = 0.$

Takeaway (Problems 7 & 8)

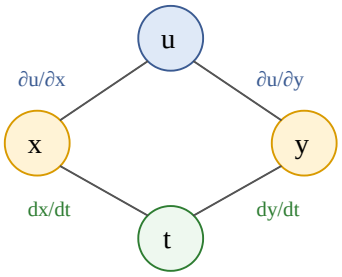
When a function depends on x, y, z only through r, introduce r and use $r_x = x/r$. It turns long calculations into short ones, and symmetry gives you the y and z versions for free.

Part 2 — Chain Rule, Total Derivative, Implicit Functions

2.1 When the inputs themselves are moving

Suppose the temperature on a plate is $u(x, y)$, and an ant walks across the plate so that its position at time t is $(x(t), y(t))$. The temperature the ant feels changes with time. How fast?

> **Big idea: changes ripple through every path**
>
> Time t affects u in two ways: through x and through y. The total rate of change is the sum of the contributions of each path:
>
> $$\frac{du}{dt} = \underbrace{\frac{\partial u}{\partial x}\frac{dx}{dt}}_{\text{effect through } x} + \underbrace{\frac{\partial u}{\partial y}\frac{dy}{dt}}_{\text{effect through } y}.$$
>
> Each path = (how sensitive u is to that variable) × (how fast that variable is changing).



A dependency tree. To get du/dt: for each path from u down to t, multiply the derivatives along it, then add the paths.

If x, y depend on two variables, say u and v (as in Problem 13), the same rule works for each of them separately, using partial derivatives:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}.$$

2.2 Implicit functions: curves given by an equation

Some curves are given by an equation like $x^3 + y^3 = 6xy$ instead of $y = \ldots$. You can't easily solve for y, but you can still find the slope dy/dx.

> **Why $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$**
>
> Write the curve as $f(x, y) = 0$. As you move along the curve, f **stays equal to 0**, so its total rate of change is 0. Using the chain rule with y depending on x:
>
> $$\frac{d}{dx} f(x, y) = f_x \cdot 1 + f_y \cdot \frac{dy}{dx} = 0 \quad \Longrightarrow \quad \frac{dy}{dx} = -\frac{f_x}{f_y}.$$
>
> In words: a step in x would change f by f_x; to stay on the curve, y must move just enough to cancel that out.

The second derivative. Differentiate $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$ once more, remembering f_x and f_y also change as we move along the curve ($\frac{d}{dx} f_x = f_{xx} + f_{xy} y'$ and $\frac{d}{dx} f_y = f_{xy} + f_{yy} y'$). Substituting $y' = -f_x / f_y$ and tidying up gives

$$\frac{d^2y}{dx^2} = -\frac{f_{xx} f_y^2 - 2 f_{xy} f_x f_y + f_{yy} f_x^2}{f_y^3}.$$

You can memorise this, or redo the derivation each time. Both are fine.

Problem 9. If $x^3 + y^3 - 3axy = 0$, find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$.

Picture

This curve is the **Folium of Descartes**, a loop shaped like a leaf. We can't write y simply in terms of x, so we use the implicit formulas from 2.2.

Step 1: Name the function. $f = x^3 + y^3 - 3axy$.
$f_x = 3x^2 - 3ay$ (treat y as constant), $f_y = 3y^2 - 3ax$ (treat x as constant).

Step 2: First derivative.

$$\frac{dy}{dx} = -\frac{3x^2 - 3ay}{3y^2 - 3ax} = \frac{ay - x^2}{y^2 - ax}.$$

We cancelled the 3s and moved the minus sign into the top.

Step 3: The second partials. $f_{xx} = 6x$, $f_{xy} = -3a$, $f_{yy} = 6y$.

Step 4: Shorthand to keep it neat. Let $p = x^2 - ay$ and $q = y^2 - ax$. Then $f_x = 3p$ and $f_y = 3q$. Plug into the formula:

$$\text{top} = 6x \cdot 9q^2 - 2(-3a)(3p)(3q) + 6y \cdot 9p^2 = 54\,(xq^2 + apq + yp^2), \qquad \text{bottom} = (3q)^3 = 27q^3.$$

Why bother with p and q?

Without shorthand the expression is several lines long and mistakes are almost certain. Short letters let you see the structure. We expand them back only at the end.

Step 5: Expand the bracket.

$$\begin{aligned} xq^2 &= x(y^2 - ax)^2 = xy^4 - 2ax^2y^2 + a^2x^3 \\ yp^2 &= y(x^2 - ay)^2 = x^4y - 2ax^2y^2 + a^2y^3 \\ apq &= a(x^2 - ay)(y^2 - ax) = ax^2y^2 - a^2x^3 - a^2y^3 + a^3xy \end{aligned}$$

Add: the a^2x^3 and a^2y^3 terms cancel, and we are left with

$$xy^4 + x^4y - 3ax^2y^2 + a^3xy = xy\,\underbrace{(x^3 + y^3 - 3axy)}_{=0 \text{ on the curve}} + a^3xy = a^3xy.$$

Why did we factor out xy?

We noticed $xy^4 + x^4y - 3ax^2y^2$ looks like the original curve multiplied by xy. Since the point is *on* the curve, that part is 0. Always check whether the original equation can be reused.

Step 6: Assemble.

$$\frac{d^2y}{dx^2} = -\frac{54a^3xy}{27(y^2 - ax)^3} = -\frac{2a^3xy}{(y^2 - ax)^3} = \frac{2a^3xy}{(ax - y^2)^3}.$$

In the last step we flipped the sign inside the cube: $(y^2 - ax)^3 = -(ax - y^2)^3$.

Problem 10. Find $\dfrac{dy}{dx}$ when $(\cos x)^y = (\sin y)^x$.

Game plan

Variables are in both the base and the exponent, which is awkward. Taking logarithms brings exponents down ($\log(a^b) = b\log a$). Then use the implicit formula.

Step 1: Take log of both sides. $y\log(\cos x) = x\log(\sin y)$.

Step 2: Move everything to one side. $f(x,y) = y\log\cos x - x\log\sin y = 0$.

Step 3: Partials.

- f_x: y is constant. $\dfrac{d}{dx}\log\cos x = \dfrac{-\sin x}{\cos x} = -\tan x$. So the first term gives $-y\tan x$. The second term $x\log\sin y$ is x times a constant, giving $\log\sin y$. Hence $f_x = -y\tan x - \log\sin y$.
- f_y: x is constant. First term gives $\log\cos x$. $\dfrac{d}{dy}\log\sin y = \dfrac{\cos y}{\sin y} = \cot y$, so the second gives $-x\cot y$. Hence $f_y = \log\cos x - x\cot y$.

Step 4:

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = \frac{y\tan x + \log\sin y}{\log\cos x - x\cot y}.$$

> **Takeaway**
> Variables in the exponent ⇒ take logs first. Equation instead of $y = \ldots$ ⇒ use $-f_x/f_y$.

Problem 11. If $u = x\log(xy)$ where $x^3 + y^3 + 3xy = 1$, find $\dfrac{du}{dx}$.

> **What's special**
> u depends on x and y, but y is tied to x by the curve equation. So when x changes, u changes *directly* and also *through* y. This is the **total derivative**:
>
> $$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dx}.$$
>
> (It is the chain rule of 2.1 with t replaced by x, and $\dfrac{dx}{dx} = 1$.)

Step 1: Partials of u. Use $\log(xy) = \log x + \log y$, so $u = x\log x + x\log y$.
$\dfrac{\partial u}{\partial x} = (\log x + 1) + \log y = 1 + \log(xy)$ (product rule on $x\log x$).
$\dfrac{\partial u}{\partial y} = x\cdot\dfrac{1}{y} = \dfrac{x}{y}$.

Step 2: dy/dx from the constraint. $f = x^3 + y^3 + 3xy - 1$: $f_x = 3x^2 + 3y$, $f_y = 3y^2 + 3x$, so $\dfrac{dy}{dx} = -\dfrac{x^2+y}{y^2+x}$.

Step 3: Combine.

$$\frac{du}{dx} = 1 + \log(xy) - \frac{x}{y}\cdot\frac{x^2+y}{y^2+x}.$$

> **Watch out**
> $\dfrac{du}{dx}$ (total) and $\dfrac{\partial u}{\partial x}$ (partial) are different! The partial ignores that y moves; the total includes it.

Problem 12. If $u = x^3 + y^3$, $x = a\cos t$, $y = b\sin t$, find $\dfrac{du}{dt}$ and verify.

> **Game plan**
> Use the chain rule (two paths: through x and through y). Then "verify" by substituting first and differentiating normally. Both methods must agree, which is a great self-check.

Step 1: Pieces. $u_x = 3x^2$, $u_y = 3y^2$, $\dfrac{dx}{dt} = -a\sin t$, $\dfrac{dy}{dt} = b\cos t$.

Step 2: Chain rule. $\dfrac{du}{dt} = 3x^2(-a\sin t) + 3y^2(b\cos t)$. Put $x = a\cos t$, $y = b\sin t$:

$$\frac{du}{dt} = -3a^3\cos^2 t\sin t + 3b^3\sin^2 t\cos t.$$

Step 3: Verify directly. $u = a^3\cos^3 t + b^3\sin^3 t$. Chain rule on $\cos^3 t$: $3\cos^2 t\cdot(-\sin t)$; on $\sin^3 t$: $3\sin^2 t\cos t$. Same answer. Verified.

> **Why learn the chain-rule way if direct substitution works?**
>
> Here substitution is easy. In harder problems (like 13 and 16), f is an unknown function, so you *can't* substitute and the chain rule is the only way.

Problem 13. If $z = f(x,y)$, $x = e^u\cos v$, $y = e^u\sin v$, show (i) $yz_u + xz_v = e^{2u}z_y$; (ii) $z_x^2 + z_y^2 = e^{-2u}(z_u^2 + z_v^2)$.

> **Setting the scene**
>
> We don't know what f is. That is the point: the identity must hold for *every* function. So we can only use the chain rule to connect the (u,v) derivatives with the (x,y) derivatives.

Step 1: Derivatives of x and y. $x_u = e^u\cos v = x$, $y_u = e^u\sin v = y$, $x_v = -e^u\sin v = -y$, $y_v = e^u\cos v = x$. Also $x^2 + y^2 = e^{2u}(\cos^2 v + \sin^2 v) = e^{2u}$.

> **Why rewrite them as x and y?**
>
> The answer is wanted in terms of x, y-derivatives, and these substitutions make everything short. Spotting that $x_u = x$ is a big simplification.

Step 2: Chain rule.

$$z_u = z_x\,x_u + z_y\,y_u = x\,z_x + y\,z_y, \qquad z_v = z_x\,x_v + z_y\,y_v = -y\,z_x + x\,z_y.$$

Step 3: Part (i).

$$y\,z_u + x\,z_v = xy\,z_x + y^2 z_y - xy\,z_x + x^2 z_y = (x^2+y^2)z_y = e^{2u}z_y.$$

The z_x terms cancel exactly.

Step 4: Part (ii). Square and add:

$$z_u^2 + z_v^2 = (xz_x + yz_y)^2 + (-yz_x + xz_y)^2.$$

Expanding: $x^2z_x^2 + 2xyz_xz_y + y^2z_y^2 \;+\; y^2z_x^2 - 2xyz_xz_y + x^2z_y^2$. The middle terms cancel, leaving $(x^2+y^2)(z_x^2 + z_y^2) = e^{2u}(z_x^2+z_y^2)$. Divide by e^{2u} to get (ii).

> **Takeaway**
>
> "Show an identity for an unknown f" ⇒ write the new derivatives in terms of the old ones with the chain rule, then do algebra.

Problem 14. $u = x^2 + 2xy - y\ln z$, $x = s + t^2$, $y = s - t^2$, $z = 2t^2$. Find $\dfrac{\partial u}{\partial s}, \dfrac{\partial u}{\partial t}$ at $(x,y,z) = (1,2,1)$.

> **Game plan**
>
> Now there are three middle variables (x,y,z) and two final ones (s,t). Each partial derivative has three paths, one through each middle variable.

Step 1: Sensitivities of u. $u_x = 2x + 2y$, $u_y = 2x - \ln z$, $u_z = -\dfrac{y}{z}$.

Step 2: How middle variables move.

With respect to s: $x_s = 1$, $y_s = 1$, $z_s = 0$ (z has no s in it).

With respect to t: $x_t = 2t$, $y_t = -2t$, $z_t = 4t$.

Step 3: Add the paths.

$$u_s = u_x(1) + u_y(1) + u_z(0) = 4x + 2y - \ln z,$$

$$u_t = (2x + 2y)(2t) + (2x - \ln z)(-2t) - \frac{y}{z}(4t) = 2t\left(2y + \ln z - \frac{2y}{z}\right).$$

Step 4: Plug in $x = 1, y = 2, z = 1$ ($\ln 1 = 0$): $u_s = 4 + 4 = 8$, $u_t = 2t(4 + 0 - 4) = 0$.

> **Note**
>
> The bracket for u_t is zero, so $u_t = 0$ no matter what t is. That's why the answer doesn't need a value of t.

Problem 15. For the curve $xy^3 - yx^3 = 6$, find the slope and tangent line at $(1, 2)$.

> **Game plan**
>
> A tangent line needs a point (given) and a slope (dy/dx at that point). Get the slope from $-f_x/f_y$, then use the point-slope form $y - y_1 = m(x - x_1)$.

Step 0: Check the point is on the curve. $1 \cdot 2^3 - 2 \cdot 1^3 = 8 - 2 = 6$. Yes.

Step 1: $f = xy^3 - yx^3 - 6$. $f_x = y^3 - 3x^2y$, $f_y = 3xy^2 - x^3$.

Step 2: Evaluate at $(1, 2)$. $f_x = 8 - 6 = 2$, $f_y = 12 - 1 = 11$. Slope $m = -\dfrac{2}{11}$.

Step 3: Tangent line. $y - 2 = -\dfrac{2}{11}(x - 1)$. Multiply by 11: $11y - 22 = -2x + 2$, i.e. $2x + 11y - 24 = 0$.

> **Meaning**
>
> Negative slope: near $(1, 2)$ the curve goes downhill from left to right, gently (for every 11 steps right, 2 steps down).

Problem 16. If $w = f(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$, show $w_r^2 + \dfrac{1}{r^2}w_\theta^2 = w_x^2 + w_y^2$.

> **Meaning**
>
> $w_x^2 + w_y^2$ is the square of the steepest slope of the surface (the size of the "gradient"). This identity says you get the same steepness whether you measure it in ordinary (x, y) coordinates or polar (r, θ) coordinates, as long as you remember that moving through angle θ at distance r covers a distance $r\,\theta$ (hence the $\frac{1}{r^2}$).

Step 1: $x_r = \cos\theta$, $y_r = \sin\theta$, $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$.

Step 2: Chain rule. $w_r = w_x\cos\theta + w_y\sin\theta$, $w_\theta = -r\sin\theta\, w_x + r\cos\theta\, w_y$, so $\dfrac{1}{r}w_\theta = -w_x\sin\theta + w_y\cos\theta$.

Step 3: Square and add.

$$w_r^2 = w_x^2\cos^2\theta + 2w_xw_y\sin\theta\cos\theta + w_y^2\sin^2\theta$$

$$\tfrac{1}{r^2}w_\theta^2 = w_x^2\sin^2\theta - 2w_xw_y\sin\theta\cos\theta + w_y^2\cos^2\theta$$

Adding, the cross terms cancel and $\sin^2\theta + \cos^2\theta = 1$ gives $w_x^2 + w_y^2$. Done.

> **Takeaway (Problems 13 & 16)**
>
> These are the same trick: chain rule, square, add, and watch the cross terms cancel thanks to $\sin^2 + \cos^2 = 1$.

Part 3 — Homogeneous Functions and Euler's Theorem

3.1 What "homogeneous" means

> **Big idea: how does the output react if you scale every input?**
>
> Take a rectangle with area $A = xy$. Double both sides: the area becomes $(2x)(2y) = 4xy$, i.e. 2^2 times bigger. Triple both: 3^2 times bigger. In general, scaling all inputs by t scales the output by t^2. We say A is **homogeneous of degree 2**.

Definition. $f(x, y)$ is homogeneous of degree n if

$$f(tx, ty) = t^n f(x, y) \quad \text{for every } t > 0.$$

Function	Replace $x \to tx,\ y \to ty$	Degree
$x^2 + 3xy$	$t^2(x^2 + 3xy)$	2
$x^3 + y^3$ over $x - y$	$t^3/t^1 = t^2$	2
$\sqrt{x^2 + y^2}$	$t\sqrt{x^2 + y^2}$	1
$\dfrac{x + y}{\sqrt{x} + \sqrt{y}}$	$t/t^{1/2} = t^{1/2}$	$\frac{1}{2}$
$x^2 + y$	$t^2x^2 + ty$, not a single power	not homogeneous

> **Quick way to find the degree**
>
> For a fraction of "nice" pieces: degree = (degree of top) − (degree of bottom). A square root halves the degree; $x^{1/3}$ has degree $\frac{1}{3}$, and so on. Every term in the top must have the same degree (and same for the bottom).

3.2 Euler's theorem and why it is true

> **Euler's theorem**
>
> If f is homogeneous of degree n, then
>
> $$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = n\,f.$$
>
> (With three variables, add $z\,f_z$ on the left.)

Why it's true (short proof). Start from $f(tx, ty) = t^n f(x, y)$ and differentiate both sides with respect to t:

- Left side, by the chain rule (inputs are tx and ty, whose t-derivatives are x and y): $x f_x(tx, ty) + y f_y(tx, ty)$.
- Right side: $nt^{n-1} f(x, y)$.

Now put $t = 1$: $x f_x + y f_y = nf$. That's it.

> **Intuition**
>
> $x f_x + y f_y$ measures "how much f changes when you stretch all inputs by a tiny percentage". For a homogeneous function, stretching by 1% changes the output by about n%, so this quantity is n times f.

3.3 The trick for inverse-trig problems

In Problems 18–22, u itself is **not** homogeneous (it is $\sin^{-1}$ or $\tan^{-1}$ of something). But the "something" is. So we:

1. Write the homogeneous part as f, e.g. $\sin u = f(x, y)$.

2. Apply Euler to f: $xf_x + yf_y = nf$.
3. Use the chain rule: $f_x = \cos u \cdot u_x$, $f_y = \cos u \cdot u_y$. Substitute and solve for $xu_x + yu_y$.

The answer is always of the form $xu_x + yu_y = G(u)$ for some function G.

3.4 The second-order formula

Formula

If $xu_x + yu_y = G(u)$, then

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = G(u)\,\big[G'(u) - 1\big].$$

Where it comes from.

1. Differentiate $xu_x + yu_y = G(u)$ w.r.t. x. Product rule on xu_x gives $u_x + xu_{xx}$; then yu_y gives yu_{xy}; right side gives $G'(u)\,u_x$:

   $$u_x + xu_{xx} + yu_{xy} = G'u_x.$$

   Multiply by x: $xu_x + x^2u_{xx} + xyu_{xy} = G'\,xu_x$.
2. Same w.r.t. y, multiply by y: $yu_y + xyu_{xy} + y^2u_{yy} = G'\,yu_y$.
3. Add: $(xu_x + yu_y) + x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = G'(xu_x + yu_y)$.
4. Replace $xu_x + yu_y$ with G: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = G'G - G = G(G' - 1)$.

Why multiply by x and y?

Because the target expression has x^2u_{xx} and y^2u_{yy}. Multiplying the first equation by x creates x^2u_{xx}, and the second by y creates y^2u_{yy}. Then adding gives the target.

Problem 17. Verify Euler's theorem for $z = \dfrac{x^{1/3} + y^{1/3}}{x^{1/2} + y^{1/2}}$.

Game plan

"Verify" = find the degree n, then compute $xz_x + yz_y$ *the long way* and check it equals nz.

Step 1: Degree. Replace $x \to tx$, $y \to ty$: top becomes $t^{1/3}(\ldots)$, bottom $t^{1/2}(\ldots)$. So $z \to t^{1/3-1/2}z = t^{-1/6}z$. Degree $n = -\frac{1}{6}$. Euler predicts $xz_x + yz_y = -\frac{1}{6}z$.

Step 2: Short names. $N = x^{1/3} + y^{1/3}$ (numerator), $D = x^{1/2} + y^{1/2}$ (denominator). Then $N_x = \frac{1}{3}x^{-2/3}$, $D_x = \frac{1}{2}x^{-1/2}$.

Step 3: Quotient rule. $z_x = \dfrac{N_xD - ND_x}{D^2} = \dfrac{\frac{1}{3}x^{-2/3}D - \frac{1}{2}x^{-1/2}N}{D^2}$. Multiply by x (recall $x \cdot x^{-2/3} = x^{1/3}$ and $x \cdot x^{-1/2} = x^{1/2}$):

$$xz_x = \frac{\frac{1}{3}x^{1/3}D - \frac{1}{2}x^{1/2}N}{D^2}, \qquad yz_y = \frac{\frac{1}{3}y^{1/3}D - \frac{1}{2}y^{1/2}N}{D^2}.$$

Step 4: Add. Notice $x^{1/3} + y^{1/3} = N$ and $x^{1/2} + y^{1/2} = D$:

$$xz_x + yz_y = \frac{\frac{1}{3}ND - \frac{1}{2}DN}{D^2} = \left(\tfrac{1}{3} - \tfrac{1}{2}\right)\frac{N}{D} = -\frac{1}{6}z.$$

Matches Euler. Verified.

Notice

The $\frac{1}{3}$ and $\frac{1}{2}$ came straight from the powers in the top and bottom. That is Euler's theorem in action: degree $= \frac{1}{3} - \frac{1}{2}$.

Problem 18. If $u = \cos^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show $xu_x + yu_y + \frac{1}{2}\cot u = 0$.

Step 1: Find the homogeneous part. $\cos u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}} = f$. Top has degree 1, bottom degree $\frac{1}{2}$, so $n = \frac{1}{2}$.

Step 2: Euler on f. $xf_x + yf_y = \frac{1}{2}f$.

Step 3: Chain rule. $f = \cos u$, so $f_x = -\sin u \cdot u_x$ and $f_y = -\sin u \cdot u_y$. Substitute:

$$-\sin u\,(xu_x + yu_y) = \tfrac{1}{2}\cos u.$$

Step 4: Solve. Divide by $-\sin u$: $xu_x + yu_y = -\frac{1}{2}\dfrac{\cos u}{\sin u} = -\frac{1}{2}\cot u$. Move it across: $xu_x + yu_y + \frac{1}{2}\cot u = 0$.

Why not differentiate u directly?

You could, but it is long and messy. Euler's theorem gives the answer in four lines, without ever computing u_x or u_y separately.

Problem 19. If $u = \sin^{-1}\left[\dfrac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right]$, show $xu_x + yu_y + zu_z + 3\tan u = 0$.

Step 1: Degree. $\sin u = f = \dfrac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$. Top: degree 1. Bottom: $\sqrt{\text{degree } 8}$ has degree 4. So $n = 1 - 4 = -3$

.

Step 2: Euler (three variables). $xf_x + yf_y + zf_z = -3f$.

Step 3: Chain rule. $f_x = \cos u \cdot u_x$, etc. So $\cos u\,(xu_x + yu_y + zu_z) = -3\sin u$.

Step 4: Divide by $\cos u$: $xu_x + yu_y + zu_z = -3\tan u$, i.e. $xu_x + yu_y + zu_z + 3\tan u = 0$.

Watch out

The degree of $\sqrt{x^8 + \ldots}$ is $8 \div 2 = 4$, not 8. A negative degree is perfectly fine: it means the function gets *smaller* when inputs are scaled up.

Problem 20. If $u = \tan^{-1}\left(\dfrac{x^3+y^3}{x-y}\right)$, prove (i) $xu_x + yu_y = \sin 2u$; (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 2\cos 3u \sin u$.

Step 1: Degree. $\tan u = f = \dfrac{x^3+y^3}{x-y}$: degree $3 - 1 = 2$.

Step 2: Euler and chain rule. $f_x = \sec^2 u \cdot u_x$, so $\sec^2 u\,(xu_x + yu_y) = 2\tan u$.

Step 3: Solve for (i). Divide by $\sec^2 u$, i.e. multiply by $\cos^2 u$:

$$xu_x + yu_y = 2\frac{\sin u}{\cos u}\cos^2 u = 2\sin u\cos u = \sin 2u.$$

(Double-angle identity: $\sin 2u = 2\sin u\cos u$.)

Step 4: Part (ii) with the formula of 3.4. $G(u) = \sin 2u$, so $G'(u) = 2\cos 2u$:

$$G(G' - 1) = \sin 2u\,(2\cos 2u - 1) = 2\sin 2u\cos 2u - \sin 2u = \sin 4u - \sin 2u.$$

(Again $2\sin A\cos A = \sin 2A$ with $A = 2u$.)

Step 5: Match the requested form. Use $\sin A - \sin B = 2\cos\dfrac{A+B}{2}\sin\dfrac{A-B}{2}$ with $A = 4u$, $B = 2u$:

$$\sin 4u - \sin 2u = 2\cos 3u \sin u.$$

Why the trig identities at the end?

The math is finished at Step 4; the identities just rewrite the answer into the form the question asked for. When you get stuck matching forms, try the double-angle and sum-to-product formulas.

Problem 21. If $u = \sin^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, prove $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = -\dfrac{\sin u \cos 2u}{4\cos^3 u}$.

Step 1: Degree. $\sin u = f$ with degree $1 - \frac{1}{2} = \frac{1}{2}$.

Step 2: First-order result. $\cos u\,(xu_x + yu_y) = \frac{1}{2}\sin u \Rightarrow G(u) = xu_x + yu_y = \frac{1}{2}\tan u$.

Step 3: G'. $G'(u) = \frac{1}{2}\sec^2 u = \dfrac{1}{2\cos^2 u}$. So

$$G' - 1 = \frac{1 - 2\cos^2 u}{2\cos^2 u} = -\frac{\cos 2u}{2\cos^2 u} \quad (\text{since } 2\cos^2 u - 1 = \cos 2u).$$

Step 4: Multiply.

$$G(G'-1) = \frac{1}{2}\cdot\frac{\sin u}{\cos u}\cdot\left(-\frac{\cos 2u}{2\cos^2 u}\right) = -\frac{\sin u\cos 2u}{4\cos^3 u}.$$

Done.

Problem 22. If $u = \sin^{-1}\sqrt{x^2+y^2}$, show (i) $xu_x + yu_y = \tan u$; (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \tan^3 u$.

Step 1: Degree. $\sin u = \sqrt{x^2+y^2}$ has degree 1 (it is the distance from the origin; doubling the point doubles the distance).

Step 2: (i). $\cos u\,(xu_x + yu_y) = 1\cdot\sin u \Rightarrow xu_x + yu_y = \tan u$.

Step 3: (ii). $G = \tan u$, $G' = \sec^2 u$, and $\sec^2 u - 1 = \tan^2 u$:

$$G(G'-1) = \tan u\cdot\tan^2 u = \tan^3 u.$$

Takeaway for Part 3 (a recipe)

1. Isolate the homogeneous piece: $\sin u$, $\cos u$ or $\tan u = f$.
2. Find its degree n (top degree minus bottom degree).
3. Euler + chain rule ⇒ $xu_x + yu_y = G(u)$.
4. Second order: $G(G'-1)$, then tidy with trig identities.

Part 4 — Taylor and Maclaurin Series

4.1 The idea: replace a hard function by a polynomial

> **Big idea**
>
> Functions like $e^x \sin y$ or $\tan^{-1}(y/x)$ are hard to calculate by hand. Polynomials (just $+$, $-$, $\times$) are easy. A **Taylor series** builds a polynomial that copies the function very closely *near one chosen point*, by matching its value, its slopes, its curvatures, and so on.

One-variable warm-up. Near $x = a$:

$$f(x) \approx \underbrace{f(a)}_{\text{same height}} + \underbrace{f'(a)(x-a)}_{\text{same slope}} + \underbrace{\frac{f''(a)}{2!}(x-a)^2}_{\text{same bending}} + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

- Keep only the first term: a flat line at the right height. Rough.
- Add the second: the tangent line. Better.
- Add the third: a parabola that bends the same way. Better still.

> **Why divide by 2!, 3!, …?**
>
> Differentiate $(x-a)^2$ twice and you get 2; differentiate $(x-a)^3$ three times and you get $3 \cdot 2 \cdot 1 = 6 = 3!$. Dividing by $k!$ cancels this, so that the k-th derivative of the polynomial at a is exactly $f^{(k)}(a)$. That is what "matching" means.

4.2 Two variables

Now we can move in x by $h = x - a$ and in y by $k = y - b$. The same idea gives:

$$f(x,y) = f + \big(hf_x + kf_y\big) + \frac{1}{2!}\big(h^2 f_{xx} + 2hk\, f_{xy} + k^2 f_{yy}\big) + \frac{1}{3!}\big(h^3 f_{xxx} + 3h^2 k\, f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\big) + \cdots$$

with all derivatives evaluated at the point (a, b).

> **Memory trick**
>
> The numbers $1, 2, 1$ and $1, 3, 3, 1$ are the binomial coefficients from $(h+k)^2$ and $(h+k)^3$ (Pascal's triangle).
>
> The k-th group is like $\dfrac{1}{k!}\left(h\dfrac{\partial}{\partial x} + k\dfrac{\partial}{\partial y}\right)^k f$.

When the point is $(0, 0)$, it is called a **Maclaurin series**, and $h = x$, $k = y$.

> **Recipe**
>
> 1. List all needed derivatives (up to the required degree).
> 2. Evaluate each at the given point. These are just numbers.
> 3. Plug into the formula and simplify.

> **Problem 23.** Expand $e^x \sin y$ in ascending powers of x and y.

"Powers of x and y" means around $(0, 0)$: a Maclaurin series. We go up to degree 3.

Step 1: Derivatives. Each x-derivative leaves e^x unchanged. Each y-derivative cycles $\sin y \to \cos y \to -\sin y \to -\cos y$. At $(0, 0)$: $e^0 = 1$, $\sin 0 = 0$, $\cos 0 = 1$.

Derivative	Formula	Value at (0,0)
f	$e^x \sin y$	0
f_x / f_y	$e^x \sin y$ / $e^x \cos y$	0 / 1
f_{xx} / f_{xy} / f_{yy}	$e^x \sin y$ / $e^x \cos y$ / $-e^x \sin y$	0 / 1 / 0
f_{xxx} / f_{xxy} / f_{xyy} / f_{yyy}	$e^x \sin y$ / $e^x \cos y$ / $-e^x \sin y$ / $-e^x \cos y$	0 / 1 / 0 / -1

Step 2: Plug in.

$$e^x \sin y = 0 + (x \cdot 0 + y \cdot 1) + \frac{1}{2}(x^2 \cdot 0 + 2xy \cdot 1 + y^2 \cdot 0) + \frac{1}{6}(x^3 \cdot 0 + 3x^2 y \cdot 1 + 3xy^2 \cdot 0 + y^3 \cdot (-1))$$
$$= y + xy + \frac{1}{2}x^2 y - \frac{1}{6}y^3 + \cdots$$

Sanity check

$e^x = 1 + x + \frac{x^2}{2} + \cdots$ and $\sin y = y - \frac{y^3}{6} + \cdots$. Multiplying: $y + xy + \frac{x^2 y}{2} - \frac{y^3}{6} + \ldots$ Same answer! This multiplying trick is a great way to check your work.

Problem 24. Expand $\tan^{-1}\dfrac{y}{x}$ about $(1,1)$ up to second-degree terms, and estimate $f(1.1, 0.9)$.

Why this is useful

Computing $\tan^{-1}(0.9/1.1)$ by hand is hard. But near $(1,1)$ we know everything ($\tan^{-1} 1 = \pi/4$). The Taylor polynomial lets us "walk" from the easy point $(1,1)$ to the nearby point $(1.1, 0.9)$.

Step 1: Value. $f(1,1) = \tan^{-1} 1 = \dfrac{\pi}{4} \approx 0.7854$.

Step 2: First derivatives. Chain rule with $\dfrac{d}{dt}\tan^{-1} t = \dfrac{1}{1+t^2}$ and $t = y/x$:

$$f_x = \frac{1}{1 + y^2/x^2} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2}, \qquad f_y = \frac{1}{1 + y^2/x^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}.$$

At $(1,1)$: $f_x = -\frac{1}{2}$, $f_y = \frac{1}{2}$.

Step 3: Second derivatives (quotient rule): $f_{xx} = \dfrac{2xy}{(x^2+y^2)^2} = \dfrac{2}{4} = \dfrac{1}{2}$, $f_{xy} = \dfrac{y^2 - x^2}{(x^2+y^2)^2} = 0$, $f_{yy} = -\dfrac{2xy}{(x^2+y^2)^2} = -\dfrac{1}{2}$.

Step 4: Build the polynomial with $h = x - 1$, $k = y - 1$:

$$f \approx \frac{\pi}{4} - \frac{1}{2}h + \frac{1}{2}k + \frac{1}{2}\left(\frac{1}{2}h^2 + 0 - \frac{1}{2}k^2\right) = \frac{\pi}{4} - \frac{h}{2} + \frac{k}{2} + \frac{h^2}{4} - \frac{k^2}{4}.$$

Step 5: Estimate. $h = 0.1$, $k = -0.1$:

$$f \approx 0.7854 - 0.05 - 0.05 + 0.0025 - 0.0025 = 0.6854.$$

Adding the third-degree terms, $-\frac{1}{12}(h^3 + 3h^2 k - 3hk^2 - k^3) = +0.0003$, gives 0.6857. The true value is $0.68573\ldots$, so the approximation is excellent.

Why it works so well

h and k are small (0.1), so h^2 is 0.01, h^3 is 0.001, and each extra degree adds a much smaller correction. That's why a few terms are enough.

Problem 25. Expand $x^2y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$.

What is being asked?

Powers of $(x-1)$ and $(y+2)$ means we centre at $x=1$, $y=-2$ (because $y+2 = y-(-2)$). The function is already a polynomial, so the series **ends** after finitely many terms and is **exact**, not just an approximation. We are just rewriting it in a new "language".

Step 1: Values at $(1,-2)$. $f = 1\cdot(-2) + 3(-2) - 2 = -10$.
$f_x = 2xy = -4$, $f_y = x^2 + 3 = 4$.
$f_{xx} = 2y = -4$, $f_{xy} = 2x = 2$, $f_{yy} = 0$.
$f_{xxy} = 2$; every other third derivative is 0, and all fourth derivatives are 0.

Step 2: Plug in ($h = x-1$, $k = y+2$):

$$f = -10 + (-4h + 4k) + \frac{1}{2}(-4h^2 + 2\cdot 2\,hk + 0) + \frac{1}{6}(3h^2k\cdot 2) = -10 - 4h + 4k - 2h^2 + 2hk + h^2k.$$

Answer: $x^2y + 3y - 2 = -10 - 4(x-1) + 4(y+2) - 2(x-1)^2 + 2(x-1)(y+2) + (x-1)^2(y+2)$.

Check it yourself

Put $x=2$, $y=0$: original $= 0+0-2 = -2$. Series: $h=1$, $k=2$: $-10-4+8-2+4+2 = -2$. ✓

Problem 26. Expand $e^x \log(1+y)$ in powers of x, y up to third degree.

Step 1: Derivatives at $(0,0)$. x-derivatives keep e^x. For y: $\log(1+y) \to \dfrac{1}{1+y} \to -\dfrac{1}{(1+y)^2} \to \dfrac{2}{(1+y)^3}$. At $y=0$ these are $0, 1, -1, 2$.

Derivative	Value	Derivative	Value
f, f_x, f_{xx}, f_{xxx}	0 (all contain $\log 1 = 0$)	f_y, f_{xy}, f_{xxy}	1
f_{yy}, f_{xyy}	-1	f_{yyy}	2

Step 2: Plug in.

$$f = y + \frac{1}{2}(2xy - y^2) + \frac{1}{6}(3x^2y - 3xy^2 + 2y^3) = y + xy - \frac{y^2}{2} + \frac{x^2y}{2} - \frac{xy^2}{2} + \frac{y^3}{3} + \cdots$$

Check by multiplying known series

$\left(1 + x + \frac{x^2}{2}\right)\left(y - \frac{y^2}{2} + \frac{y^3}{3}\right)$ keeping terms of total degree ≤ 3 gives exactly the same thing. Note the y^3 term is $+\frac{1}{3}y^3$ (the printed answer key has a sign typo).

Part 5 — Maxima and Minima

5.1 Step one: where the ground is flat

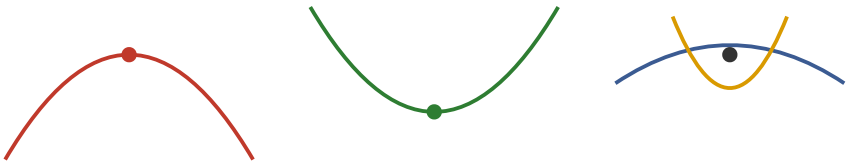
Big idea

At the very top of a hill, or the bottom of a valley, the ground is **flat**: it isn't sloping east-west or north-south. So at a maximum or minimum,

$$f_x = 0 \quad \text{and} \quad f_y = 0.$$

Points where this happens are called **stationary** (or critical) points. They are the *candidates*.

But flat ground isn't always a peak or a valley. There is a third possibility: a **saddle point**, like the middle of a horse's saddle or a Pringles chip. It curves up in one direction and down in another.



Three kinds of flat point. The second-derivative test tells them apart.

5.2 Step two: the second-derivative test

At a stationary point, let $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$.

Condition	Conclusion
$rt - s^2 > 0$ and $r < 0$	Maximum
$rt - s^2 > 0$ and $r > 0$	Minimum
$rt - s^2 < 0$	Saddle point (neither)
$rt - s^2 = 0$	Test fails; investigate another way

Why does $rt - s^2$ decide it?

Near the flat point, Taylor's formula (Part 4) says the change in height for a small step (h, k) is about

$$\Delta f \approx \tfrac{1}{2}\left(rh^2 + 2shk + tk^2\right).$$

(The first-order terms vanish because $f_x = f_y = 0$.) Complete the square, just like with quadratics in school:

$$rh^2 + 2shk + tk^2 = r\left(h + \frac{s}{r}k\right)^2 + \frac{rt - s^2}{r}\,k^2.$$

- If $rt - s^2 > 0$, both pieces have the **same sign as** r. If $r < 0$, every step goes *down*: a maximum. If $r > 0$, every step goes *up*: a minimum.
- If $rt - s^2 < 0$, the two pieces have opposite signs, so some steps go up and others go down: a saddle.

Problem 27. Test $f(x, y) = x^3y^2(6 - x - y)$ for maxima and minima (away from the origin).

Step 1: Expand. $f = 6x^3y^2 - x^4y^2 - x^3y^3$. Expanding makes differentiation simpler (just the power rule).

Step 2: First derivatives, factored.

$$f_x = 18x^2y^2 - 4x^3y^2 - 3x^2y^3 = x^2y^2(18 - 4x - 3y),$$
$$f_y = 12x^3y - 2x^4y - 3x^3y^2 = x^3y(12 - 2x - 3y).$$

> **Why factor?**
> A product is zero when one of its factors is zero. Factoring shows every way $f_x = 0$ can happen. The factors x^2y^2 and x^3y vanish on the axes ($x = 0$ or $y = 0$), which the question excludes. So the brackets must vanish.

Step 3: Solve. $4x + 3y = 18$ and $2x + 3y = 12$. Subtract the second from the first: $2x = 6$, so $x = 3$, then $3y = 12 - 6 = 6$, $y = 2$. Stationary point $(3, 2)$.

Step 4: Second derivatives at $(3, 2)$.

- $r = f_{xx} = 36xy^2 - 12x^2y^2 - 6xy^3 = 432 - 432 - 144 = -144$
- $s = f_{xy} = 36x^2y - 8x^3y - 9x^2y^2 = 648 - 432 - 324 = -108$
- $t = f_{yy} = 12x^3 - 2x^4 - 6x^3y = 324 - 162 - 324 = -162$

Step 5: Test. $rt - s^2 = 23328 - 11664 = 11664 > 0$ and $r = -144 < 0$. So it is a **maximum**.

Step 6: Value. $f(3, 2) = 27 \cdot 4 \cdot (6 - 3 - 2) = 108$.

Problem 28. Examine $f(x, y) = \sin x + \sin y + \sin(x + y)$ for maxima and minima.

Step 1: Flat points. $f_x = \cos x + \cos(x + y) = 0$ and $f_y = \cos y + \cos(x + y) = 0$.

Step 2: Subtract them. $\cos x - \cos y = 0$, so $\cos x = \cos y$. Within the usual range $0 < x, y < \pi$ this means $x = y$.

> **Why subtract?**
> Both equations contain the same awkward term $\cos(x + y)$. Subtracting removes it and leaves a simple relation between x and y.

Step 3: Substitute $y = x$. $\cos x + \cos 2x = 0$. Use $\cos 2x = 2\cos^2 x - 1$:

$$2\cos^2 x + \cos x - 1 = 0 \;\Rightarrow\; (2\cos x - 1)(\cos x + 1) = 0.$$

This is just a quadratic in $c = \cos x$: $2c^2 + c - 1 = 0$. So $\cos x = \frac{1}{2}$ ($x = \pi/3$) or $\cos x = -1$ ($x = \pi$).

Step 4: Second derivatives. $r = -\sin x - \sin(x + y)$, $s = -\sin(x + y)$, $t = -\sin y - \sin(x + y)$.

Step 5: At $(\pi/3, \pi/3)$. $\sin \frac{\pi}{3} = \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$. So $r = t = -\sqrt{3}$, $s = -\frac{\sqrt{3}}{2}$. Then $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4} > 0$ and $r < 0$: **maximum**, with value $3 \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2} \approx 2.598$.

Step 6: At (π, π). All of r, s, t are 0, so the test can't decide. Looking closer (with $x = \pi + h$, $y = \pi + k$), $f \approx -\frac{1}{2}hk(h + k)$, which is positive for some small steps and negative for others. So it is neither a max nor a min.

> **Sense check**
> Each sine is at most 1, so $f \leq 3$. Our maximum 2.598 is below 3, which makes sense: you can't make all three sines equal to 1 at once.

Part 6 — Lagrange's Method of Multipliers

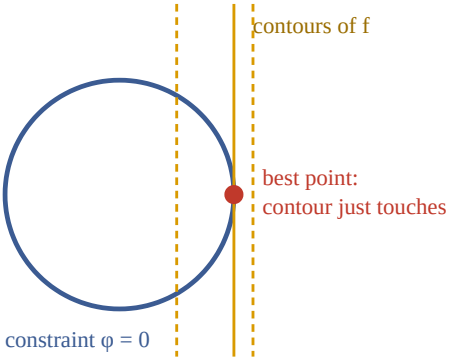
6.1 Optimising with a rule you must obey

Sometimes you want the largest or smallest value of f, but you are **not allowed** to go anywhere. You must stay on a curve or surface $\phi = 0$ (the **constraint**). Examples: the closest point *on a sphere*; the largest box that *fits inside* a sphere.

Big idea: touching, not crossing

Draw the contour lines of f (lines of equal value, like on a map) and the constraint curve. Walk along the constraint. While your path *crosses* contour lines, f is still increasing or decreasing, so you're not at the best point yet. The best point is where the path just **touches** a contour line without crossing it. There, the two curves are tangent, so their "steepest-direction" arrows (gradients) point along the same line:

$$\nabla f = -\lambda \nabla \phi \quad \text{for some number } \lambda.$$



The contour line through the best point is tangent to the constraint curve.

6.2 The recipe

1. Build $F = f + \lambda\phi$ (λ is an unknown helper number called the **multiplier**).
2. Set $F_x = 0$, $F_y = 0$, $F_z = 0$. This is exactly the "arrows are parallel" condition.
3. Solve these together with $\phi = 0$.
4. Evaluate f at the solutions; the biggest is the max, the smallest is the min.

Why add $\lambda\phi$?

On the constraint, $\phi = 0$, so adding $\lambda\phi$ doesn't change f at all. It is a clever way to turn "optimise f while staying on $\phi = 0$" into "find where all partial derivatives of F are zero", which we already know how to do.

Problem 29. Find the maximum and minimum distances of the point $(3, 4, 12)$ from the sphere $x^2 + y^2 + z^2 = 1$.

Intuition first

The sphere has centre $O = (0, 0, 0)$ and radius 1. The point $A = (3, 4, 12)$ is at distance $\sqrt{9 + 16 + 144} = \sqrt{169} = 13$ from O. Picture the straight line from A through O: the nearest point of the sphere is where this line enters it (distance $13 - 1 = 12$), and the farthest is where it exits on the far side (distance $13 + 1 = 14$). Lagrange's method should confirm this.

Step 1: What to optimise. Distance has a square root, which is annoying. Minimising the distance is the same as minimising its square, so use

$$f = (x-3)^2 + (y-4)^2 + (z-12)^2, \qquad \phi = x^2 + y^2 + z^2 - 1 = 0.$$

> **Why is squaring allowed?**
> Distances are positive, and for positive numbers, bigger number $\Leftrightarrow$ bigger square. So the max/min points are the same.

Step 2: Set up and differentiate. $F = f + \lambda\phi$.

$$F_x = 2(x-3) + 2\lambda x = 0 \;\Rightarrow\; x(1+\lambda) = 3 \;\Rightarrow\; x = \frac{3}{1+\lambda}.$$

Likewise $y = \dfrac{4}{1+\lambda}$, $z = \dfrac{12}{1+\lambda}$.

Step 3: Use the constraint. Put these into $x^2 + y^2 + z^2 = 1$:

$$\frac{9 + 16 + 144}{(1+\lambda)^2} = 1 \;\Rightarrow\; (1+\lambda)^2 = 169 \;\Rightarrow\; 1 + \lambda = \pm 13.$$

Step 4: The two points.
$1 + \lambda = 13$: $P = \left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$ (on the same side as A).
$1 + \lambda = -13$: $Q = \left(-\frac{3}{13}, -\frac{4}{13}, -\frac{12}{13}\right)$ (opposite side).

Step 5: Distances. AP: differences are $3 - \frac{3}{13} = \frac{36}{13}, \frac{48}{13}, \frac{144}{13}$, which are $\frac{12}{13} \times (3, 4, 12)$. So $AP = \frac{12}{13} \times 13 = 12$.
AQ: differences are $\frac{14}{13} \times (3, 4, 12)$, so $AQ = \frac{14}{13} \times 13 = 14$.

> **Result**
> Minimum distance 12 (at P), maximum 14 (at Q), exactly what the picture predicted.

Problem 30. Show that the rectangular box of maximum volume inside a given sphere is a cube.

> **Intuition first**
> A long thin box wastes space; a flat box wastes space. The most "balanced" shape should fill the round sphere best, and the most balanced box is a cube. Let's prove it.

Step 1: Describe the box. Put the sphere's centre at the origin, radius a: $x^2 + y^2 + z^2 = a^2$. By symmetry the best box is centred at the origin too. Its corner touching the sphere is (x, y, z), so its edges are $2x$, $2y$, $2z$ and its volume is

$$V = (2x)(2y)(2z) = 8xyz, \qquad \text{with } \phi = x^2 + y^2 + z^2 - a^2 = 0.$$

> **Why does the corner lie on the sphere?**
> If the corner were strictly inside, we could enlarge the box a little and get more volume. So the biggest box must have its corners touching the sphere.

Step 2: Lagrange. $F = 8xyz + \lambda(x^2 + y^2 + z^2 - a^2)$:

$$F_x = 8yz + 2\lambda x = 0, \qquad F_y = 8xz + 2\lambda y = 0, \qquad F_z = 8xy + 2\lambda z = 0.$$

Step 3: A neat trick. Multiply the first equation by x, the second by y, the third by z:

$$8xyz = -2\lambda x^2, \qquad 8xyz = -2\lambda y^2, \qquad 8xyz = -2\lambda z^2.$$

The left sides are all the same, so $\lambda x^2 = \lambda y^2 = \lambda z^2$. Since the box has positive volume, $\lambda \neq 0$, hence $x^2 = y^2 = z^2$, so $x = y = z$.

Why multiply by x, y, z?

It makes the same expression $8xyz$ appear in all three equations, so they can be compared directly. This is a standard trick whenever the equations are "symmetric".

Step 4: Size. $3x^2 = a^2 \Rightarrow x = y = z = \dfrac{a}{\sqrt{3}}$. All edges equal $\dfrac{2a}{\sqrt{3}}$: the box is a **cube**, with volume $\dfrac{8a^3}{3\sqrt{3}}$.

Step 5: Why it's a maximum (not a minimum). If any edge shrinks to 0, the volume is 0. The volume is positive in between, so it must reach a highest value somewhere in the middle, and the only candidate we found is the cube.

Final summary: which tool for which question?

The question says…	Use
"Find $\partial u/\partial x$"	Freeze other variables; ordinary rules + chain rule.
Inputs depend on another variable ($x(t)$, $y(t)$, or $x(u,v)\ldots$)	Chain rule: add up every path.
Curve given as an equation $f(x,y)=0$	$dy/dx = -f_x/f_y$.
$xu_x + yu_y$ appears; function is a ratio of "same-degree" pieces	Euler's theorem (and $G(G'-1)$ for second order).
"Expand in powers of…", "approximate"	Taylor/Maclaurin series.
"Maxima and minima", no restriction	$f_x = f_y = 0$, then test $rt - s^2$ and r.
"Max/min subject to…", "on the sphere", "inscribed in"	Lagrange multipliers.